

Name: _____

Date: _____

Class: _____

Element H: Prototype Testing and Data Collection Plan

Testing Details (H1, H2, H3, H4)

	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1	The SM58 Mic transmitter must fit in the slot (High priority)	1) Hold the mic head with the open end facing upwards 2) Take SM58 mic transmitter and put it in the slot 3) Note any observations about spacing of the transmitter in the slot 4) Take 3 seconds to gently shake the head, measuring whether or not the transmitter is fit securely in the slot	a. Insert transmitter (3 trials) b. Note observations (3 trials) c. Shake head part (3 trials)	Transmitter may: – rattle in slot – fall out – shift position – fit securely – stay inside	Fail: Transmitter does not fit securely in slot Pass: Transmitter fits securely in slot Goal: Transmitter can fit securely in slot, and does not rattle or have room to move.
2	Authenticity test (High priority)	1) Construct microphone assembly 2) Find 5 people 3) Ask them how authentic the microphone looks on a scale of 1-5 4) Record observations	A. Assemble microphone # trials: 1 B: Ask for feedback from people # trials: 5	Feedback may: -- Compliment microphone on it's authenticity by giving a high rating -- Be able to recognize it but not agree that it is extremely authentic -- Completely annihilate any thread of	Fail: 3 or more people do not think it is authentic Pass: 3 or more people think it is authentic Goal: 5/5 people think it is authentic

				authenticity in the design by not recognizing it as a microphone	
3	Sound quality is satisfactory (High priority)	1: Follow steps of 1st test (putting transmitter in head) 2: Follow steps of 2nd test (putting head [transmitter inside this time] on holder) 3: Shake microphone to determine whether or not microphone transmitter picks up a rattling noise 4: Speak into the microphone, and determine the quality of the sound that is produced. 5: Record sound bytes 6: Play back sound bytes and determine quality of sound	A: Assemble microphone # trials: 3 B: Shake microphone # trials: 5 C: Speak into microphone # trials: 5	Sound may: --produce a rattle noise -- not produce a rattle noise -- create a crisp sound of the voice -- create a mediocre sound of the voice	Fail: Rattle noise produced and poor sound quality of voice picked up Pass: No rattle noise is detected, and sound quality of voice is clear Goal: Sound quality of voice produced is clear and sounds good, and no rattling noise is made by the transmitter in the holder.
4	Mic Head stays on holder (velcro design) (High priority)	1: Hold holder with velcro facing upwards 2: Hold head with top hole facing upwards and velcro facing downwards. 3: Turn head over, place transmitter inside, then connect the velcro of the head and the holder (be sure to center head on holder). 4; Shake microphone up and down in a hammer motion	A: Assemble microphone # trials; 5 B: Shake Microphone # trials: 10	Head may: --Fly off and be sent across the room -- Slightly disconnect from holder -- Stay securely on holder	Fail: Mic head flies off of holder with transmitter inside and causes damage to people or objects Pass: Head stays on holder securely Goal: Head stays on holder and fits transmitter inside

					securely.
5	Mic Holder Drop Test (Medium priority)	1: Hold mic holder out at arms length 2: drop mic holder to the ground 3: observe reaction	A: Perform Drop test # of trials: 5	Mic Holder may: --fall apart --break into pieces --chip off a few pieces --be completely intact	Fail: Mic holder breaks and is permanently inoperable Pass: mic holder falls apart but is intact Goal: Mic holder is completely fine after drop test

Feedback From Field Experts (H5)

GREAT PRESENTATION

Walpole Engineers Design-a-Thon - Judge's Feedback

Judge: *Mr O* Team/Project Title: *Z Rampage*

Proficient (3)	Developing (2)	Novice (1)	Score
Learning Goal: Identify and describe a problem that can be solved with a potentially new product or process. - Can you explain what the problem is and the process you used to come to that conclusion? - What kind of input/feedback did your primary stakeholder provide for the problem and/or design requirements? - How did empathy play a role in determining the problem and design requirements? - How did you decide which design requirement was the most important to focus on?			
Problem definition and justification are clear and detailed.	Problem definition and justification are somewhat clear and objective.	Problem definition and justification are stated but unclear.	<i>3</i>
Design requirements are consistently objective and measurable and highly likely to lead to a viable solution.	Design requirements are generally objective and measurable.	Design requirements are insufficiently measurable and have little possibility of leading to a viable solution.	<i>3</i>
Learning Goal: Plan and conduct research by gathering relevant and credible data, facts, and information. - What evidence is there that your current design is based on your observations and interview of your primary stakeholder? - Where did you draw inspiration from for your current design? - What has your primary stakeholder already tried? How did you apply this information to your current design?			
The analysis of past and current attempts to solve the problem — including both strengths and shortcomings — is consistently clear, detailed, and supported by relevant data.	The analysis of past and current attempts to solve the problem — including both strengths and shortcomings — is generally clear and contains some detail and relevant supporting data.	The analysis of past and current attempts to solve the problem — including both strengths and shortcomings — is vague and is missing relevant details and/or relevant supporting data.	<i>3</i> ALL SHOWN IN STEPS IN DEVELOPMENT
Learning Goal: Articulate appropriate STEM practices and principles in the design. What kind of research did you do to learn more about the problem and how did you apply that to your current design?			
The proposed solution is well substantiated with STEM content, principles and practices.	The proposed solution is partially substantiated with STEM content, principles and practices.	The proposed solution is lacking substantiation with STEM content, principles and practices, though an attempt was made.	<i>3</i>
Learning Goal: Evaluate solution alternatives and select a final design by considering assumptions, trade offs, criteria, and constraints. - What other ideas you came up with during the brainstorming phase? - How did you make the decision to focus on your current design? - Can you walk through all the different phases/versions that your current design/prototype went through? - What did you learn or take away from all the previous versions or attempts?			

WELL DOCUMENTED STEPS

- Do you have any pictures or physical models of your previous or early attempts? - What do you really like about your current design/prototype? - What are you looking forward to improving in the next version?			
The design solution chosen is well justified and satisfies all design requirements. Strengths and weaknesses are comprehensively noted and explained well.	Strengths and weaknesses of the design solution are noted and explained.	The discussion lacks analysis related to strengths and weaknesses of the solution in meeting the design requirements.	<i>3</i>
Learning Goal: Create a prototype. - How did you make your current prototype? - What challenges did you face, and how did you overcome them? - What do you really like about your current design/prototype? - What are you looking forward to improving in the next version? - Were there any limitations or constraints that impacted what you were able to build?			
The prototype is constructed with enough detail to assure that objective data on all or nearly all design requirements could be determined.	The prototype is constructed with enough detail to assure that objective data on some design requirements could be determined.	The prototype is not constructed with enough detail to assure that objective data on at least one design requirement could be determined.	<i>3</i> LOTS OF GREAT PROTO TYPES
Learning Goal: Create and implement a Test Plan to evaluate the performance of the design solution. - What have you learned or what are you hoping to learn from testing your prototype today? - Who is your intended user and how did you consider their input in your Test Plan? - Can you measure any quantifiable constraints from your Design Brief? If so, what is an acceptable measure or result?			
The prototype testing plan addresses all or nearly all of the high priority design requirements.	The prototype testing plan addresses many of the high priority design requirements.	The prototype testing plan addresses a few of the high priority design requirements.	<i>3</i>
Learning Goal: Use various engineering communication methods.			
Student fully communicates key aspects of a design through the creation of written documents and/or presentation that addresses all prompts	Student partially communicates key aspects of a design through the creation of written documents and/or presentation that addresses all prompts	Student minimally communicates key aspects of a design through the creation of written documents and/or presentation that addresses all prompts	<i>3</i>
Student is fully able to effectively communicate important details of an object through sketching	Student is partially able to effectively communicate important details of an object through sketching	Student is minimally able to effectively communicate important details of an object through sketching	<i>3</i>
Judge's Comments: <i>WELL PREPARED - GREAT DEMO SPEECHES</i> <i>EXCELLENT COMMUNICATION W/ ME AS THEY DESIGNED</i>			

- Transcript: "Kyle- Intro= Very good. Sean- the process. Matt- Testing. Add balance to it and not top heavy"

Walpole Engineers Design-a-Thon – Judge's Feedback

Judge: Ben Love Team/Project Title: Team Nanpage – Sean, Kyle, Matt, Mic Hecker

Proficient (3)	Developing (2)	Novice (1)	Score
Learning Goal: Identify and describe a problem that can be solved with a potentially new product or process. - Can you explain what the problem is and the process you used to come to that conclusion? - What kind of input/feedback did your primary stakeholder provide for the problem and/or design requirements? - How did empathy play a role in determining the problem and design requirements? - How did you decide which design requirement was the most important to focus on?			
Problem definition and justification are clear and detailed	Problem definition and justification are somewhat clear and objective	Problem definition and justification are stated but unclear	3
Design requirements are consistently objective and measurable and highly likely to lead to a viable solution.	Design requirements are generally objective and measurable.	Design requirements are insufficiently measurable and have little possibility of leading to a viable solution	3
Learning Goal: Plan and conduct research by gathering relevant and credible data, facts, and information. - What evidence is there that your current design is based on your observations and interview of your primary stakeholder? - Where did you draw inspiration from for your current design? - What has your primary stakeholder already tried? How did you apply this information to your current design?			
The analysis of past and current attempts to solve the problem — including both strengths and shortcomings — is consistently clear, detailed, and supported by relevant data.	The analysis of past and current attempts to solve the problem — including both strengths and shortcomings — is generally clear and contains some detail and relevant supporting data.	The analysis of past and current attempts to solve the problem — including both strengths and shortcomings — is vague and is missing relevant details and/or relevant supporting data.	3
Learning Goal: Articulate appropriate STEM practices and principles in the design. What kind of research did you do to learn more about the problem and how did you apply that to your current design?			
The proposed solution is well substantiated with STEM content, principles and practices.	The proposed solution is partially substantiated with STEM content, principles and practices.	The proposed solution is lacking substantiation with STEM content, principles and practices, though an attempt was made.	3
Learning Goal: Evaluate solution alternatives and select a final design by considering assumptions, trade offs, criteria, and constraints. - What other ideas you came up with during the brainstorming phase? - How did you make the decision to focus on your current design? - Can you walk through all the different phases/versions that your current design/prototype went through? - What did you learn or take away from all the previous versions or attempts?			

Revised Testing Details That Incorporate Field Expert Feedback (H5)

Our testing plan was approved by two field experts -- Mr. O'Farrell and Professor Love -- and they both agreed that our testing plan was perfectly fine as it was.

Learning Goal: Create and implement a Test Plan to evaluate the performance of the design solution.

- What have you learned or what are you hoping to learn from testing your prototype today?
- Who is your intended user and how did you consider their input in your Test Plan?
- Can you measure any quantifiable constraints from your Design Brief? If so, what is an acceptable measure or result?

The prototype testing plan addresses all or nearly all of the high priority design requirements	The prototype testing plan addresses many of the high priority design requirements.	The prototype testing plan addresses a few of the high priority design requirements.	3
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Learning Goal: Create and implement a Test Plan to evaluate the performance of the design solution.

- What have you learned or what are you hoping to learn from testing your prototype today?
- Who is your intended user and how did you consider their input in your Test Plan?
- Can you measure any quantifiable constraints from your Design Brief? If so, what is an acceptable measure or result?

The prototype testing plan addresses all or nearly all of the high priority design requirements	The prototype testing plan addresses many of the high priority design requirements.	The prototype testing plan addresses a few of the high priority design requirements.	3
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